

Can Undemocratic Learning Be Reversed? A Two-Wave Laboratory Experiment on Institutional Capacity and Executive Delegation

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Working proposal — July 19, 2026

Abstract

Why do citizens who accept unconstrained rule under adverse conditions later return to democratic institutions? I study *undemocratic reversal*: a within-person shift from executive delegation back to constrained collective government after the structural justification for delegation disappears. The project uses a two-wave, incentivized laboratory panel conducted in synchronized sessions one week apart. In each wave, participants play ten rounds of a five-person institutional-choice public-good game. Before every allocation decision, each participant casts a secret ballot between a constrained collective procedure and an executive-delegation procedure; the majority choice binds the group. Under delegation, an elected executive can impose equal contributions and privately divert resources. Wave 1 gives delegation a productive advantage over ordinary institutions. In Wave 2, ten-person matching pools are randomly assigned either to a structural-recovery condition, which removes that advantage, or to a persistence condition, which retains it. Anonymous rematching, payoff-relevant votes, real distributive consequences, costly information, and repeated interaction make the design recognizably economic while preserving the original project's learning, memory, and process-tracing objectives. The primary outcome is whether a participant who chose delegation in Wave 1's final round chooses the constrained procedure in Wave 2's final round. Pre-information memory, costly search, belief updating, all twenty ballots, contributions, executive decisions, and realized payoffs reveal how that reversal occurs.

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1 Research Question and Contribution

Research on democratic backsliding shows that citizens sometimes trade institutional safeguards for policy congruence, material performance, or protection from an opposing camp (Fossati, Muhtadi, and Warburton 2022; Kingzette et al. 2021; Simonovits, McCoy, and Littvay 2022; Saikkonen and Christensen 2023). The harder question is dynamic: after citizens have learned that unconstrained authority is useful, what makes them abandon it? This proposal asks:

When the structural conditions that made executive discretion instrumentally attractive improve, do citizens return to constrained collective government, and through what learning and memory processes?

I call this behavioral transition *undemocratic reversal*. It is not simply low support for an undemocratic alternative, nor a between-person difference in democratic attitudes. It is an observed within-person change: a participant initially endorses an institution that concentrates authority and weakens ex ante control, then later chooses an institution that restores collective choice and constraints. The concept therefore requires a panel structure and at least two consequential institutional choices by the same person.

The design contributes in three ways. First, it turns democratic constraint into a set of payoff-relevant rules governing interaction. Participants do not merely rate political descriptions. Their ballots select who controls endowments, how a public good is produced, and whether an executive may extract a private rent. This embeds democratic trade-offs in the classical experimental-economics traditions of public-good provision, delegation, voting, and rent extraction (Hamman, Weber, and Woon 2011).

Second, it identifies a structural cause of reversal. All participants encounter low ordinary institutional capacity in Wave 1. Before Wave 2, matched pools are randomly assigned either to recovered capacity or to persistent incapacity. The executive institution itself does not improve or deteriorate. Consequently, the treatment changes the instrumental case for delegation rather than supplying a moral appeal to democracy.

Third, the experiment observes the mechanism rather than only its endpoint. Participants choose whether to buy information, state probabilistic beliefs before and after search, report what they remember after a one-week delay, vote in every round, and experience endogenous decisions by other subjects. The resulting record distinguishes durable learning from information retrieval, strategic adaptation, and repeated-game dynamics.

2 Theory

2.1 Conditional tolerance of unconstrained authority

Citizens need not reject democratic constraints in principle to support an unconstrained executive in practice. They may instead accept discretion conditionally, when ordinary institutions appear incapable of producing valued outcomes. This logic is compatible with evidence that electoral and material considerations can outweigh checks on executives and that citizens interpret democratic violations through partisan and performance-based trade-offs (Kingzette et al. 2021; Saikkonen and Christensen 2023; Wunsch, Jacob, and Derksen 2025). It also clarifies why democratic tolerance can be learned: repeated evidence of institutional failure changes the expected payoff from retaining constraints.

In the experiment, the constrained collective procedure (D) protects individual control. Each member chooses a contribution, benefits are shared equally, and no single person can appropriate

the public account. Yet its low Wave-1 productivity makes underprovision costly. The executive-delegation procedure (A) solves the coordination problem by allowing one person to impose equal contributions and by supplying a more productive implementation technology. It also permits that executive to divert part of the account. A participant can therefore support A without expressing a general taste for autocracy: under weak ordinary capacity, delegation can be a rational response to institutional performance.

H1: Crisis endorsement. In Wave 1, participants will increasingly support executive delegation when their experience indicates that its productive advantage outweighs the risks of executive extraction.

2.2 Structural updating and democratic reversal

The central theoretical claim is conditional. If support for unconstrained authority was learned from structural performance, then credible structural improvement should alter it. The Wave-2 recovery treatment raises the constrained procedure’s productivity to the executive procedure’s level. Recovery removes the executive’s technological advantage but leaves its discretion to extract rents unchanged. A participant who understands this change should update the relative value of the two institutions.

The persistence condition supplies the counterfactual. Participants return after the same delay, complete the same memory task, face the same information prices, and play the same number of new rounds; only ordinary institutional capacity differs. A treatment-control contrast therefore separates recovery-induced change from generic maturation, regression to the mean, fatigue, or simply playing for a second day.

H2: Structural updating. Participants assigned to recovery will report a larger increase in perceived ordinary institutional capacity and a larger reduction in the perceived instrumental need for delegation than participants assigned to persistence.

H3: Democratic reversal. Among participants who choose A in Wave 1’s final round, assignment to recovery will increase the probability of choosing D in Wave 2’s final round.

H4: Behavioral mechanism. Recovery will change the path of institutional choice across Wave 2: support for D should rise as participants experience its restored productivity, while executive extraction should make continued delegation less attractive.

2.3 Memory and costly information

Structural change cannot affect choice if it is neither encoded nor retrieved. Political learning may rest on a durable running evaluation or on specific considerations accessible when a decision is made. A one-week gap creates a meaningful test: before Wave-2 information is displayed, participants freely recall the prior environment, recognize its capacity level, recall their own final ballot, and state a prior probability that capacity recovered.

Information is then costly. Every participant begins the panel with a 24-point information endowment. Opening a full report costs four points, and unspent points are paid at the end. Headlines are neutral and do not disclose treatment status. Search therefore reveals a consequential demand for evidence rather than costless clicking. The same six report topics appear in each

condition; condition-specific facts appear only after purchase. Participants state the probability of recovery again after search.

Information acquisition is endogenous, not itself randomized. It is a process measure: treatment effects on the final outcome will be reported both overall and descriptively by search behavior, but search will not be treated as an exogenous mediator without additional assumptions.

H5: Information and memory. More accurate Wave-1 memory and greater acquisition of diagnostic Wave-2 information will predict more accurate recovery beliefs and a stronger alignment between final institutional choice and the realized environment.

3 Experimental Design

3.1 Overview

The study consists of two synchronized laboratory visits scheduled seven days apart. Each visit is designed for 55–65 minutes. Participants remain in stable, ten-person matching pools across waves, and treatment is randomized at the pool level. Within each pool, the software anonymously rematches participants into two five-person groups before every paid round. Thus, participants interact repeatedly within a treatment-consistent local population but cannot build a reliable bilateral reputation.

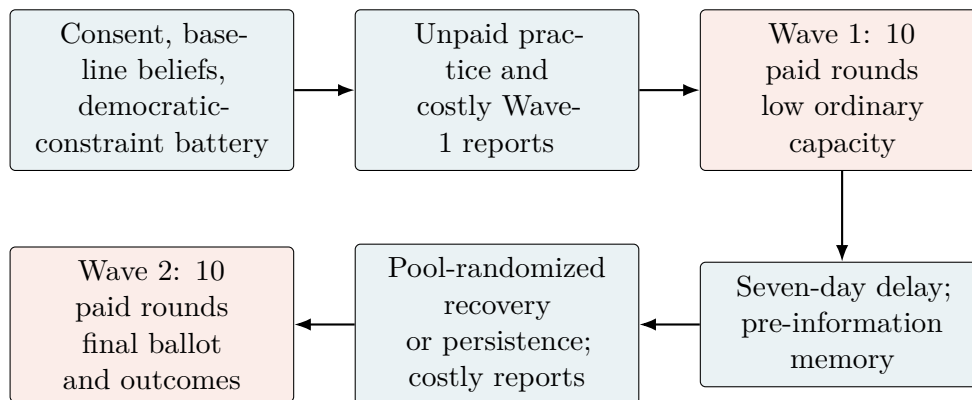


Figure 1: Two-wave experimental sequence. Every paid round contains a secret institutional ballot, an institution-contingent allocation decision, feedback, and anonymous rematching.

The ten rounds per visit solve the original design’s brevity problem while adding theoretically useful behavior. They permit participants to experience both institutions, observe executive choices, and reveal whether reversal is immediate, gradual, or unstable. The primary endpoint remains the final ballot in each wave; earlier rounds process-trace the trajectory and do not multiply the number of independent treatment units.

3.2 The institutional-choice public-good game

Each round begins with a 20-point endowment for each of five group members. Every member privately votes between the constrained collective procedure (D) and executive delegation (A). A simple majority of at least three votes selects the binding institution. Individual votes remain secret; the group sees the aggregate vote and the selected rule.

Constrained collective procedure (D). Each member chooses a contribution c_i of 0–20 points. Let m_{gw} denote ordinary institutional capacity for matching pool g in wave w . Player i earns

$$\pi_{igwr}^D = 20 - c_{igwr} + \frac{m_{gw}}{5} \sum_{j=1}^5 c_{jgwr}. \quad (1)$$

The marginal private return is below one in both capacity states, preserving the social dilemma, while the group return exceeds one. In Wave 1, $m_{g1} = 1.50$ for everyone. In Wave 2, $m_{g2} = 2.50$ under recovery and $m_{g2} = 1.50$ under persistence.

Executive-delegation procedure (A). One group member is selected as executive. Selection is balanced toward members who have served least often, with random resolution of ties. The executive chooses an equal mandatory contribution $t \in \{0, \dots, 20\}$ and a private transfer $\rho \in \{0, \dots, 20\}$, subject to $\rho \leq 5t$. The public account equals $5t - \rho$ and always uses multiplier 2.50. A nonexecutive earns

$$\pi_{igwr}^A = 20 - t + \frac{2.50}{5}(5t - \rho), \quad (2)$$

while the executive earns

$$\pi_{igwr}^{A,E} = 20 - t + \frac{2.50}{5}(5t - \rho) + \rho. \quad (3)$$

All members observe the imposed contribution, private transfer, public return, and their own payoff after the round. Delegation can coordinate provision, but it creates a distributive conflict grounded in another participant’s decision.

Table 1: Implemented design parameters

Feature	Implemented value
Laboratory visits	Two synchronized visits, seven days apart
Paid rounds	10 per wave (20 total), preceded by unpaid practice
Group and matching pool	5-person groups; 10-person treatment-matched pools
Rematching	Anonymous rematching within pool before every paid round
Round endowment	20 points per participant
Constrained multiplier	1.50 in Wave 1; 2.50 recovery / 1.50 persistence in Wave 2
Executive multiplier	2.50 in both waves and conditions
Executive private transfer	0–20 points, limited by total collected
Information endowment	24 points across the panel; 4 points per opened report
Performance payment	One randomly selected game round from each wave
Exchange rate	EUR 0.20 per point
Additional incentives	Unspent information points; 25-point completion bonus; 0–5 belief bonus
Show-up payment	EUR 10.00

The random-round rule limits wealth effects, hedging across rounds, and excessive payment while preserving incentives in every decision. The belief bonus uses a quadratic scoring rule for the post-search probability assessment.

3.3 Wave 1: acquiring undemocratic tolerance

After consent, participants receive common instructions, complete baseline capacity and risk beliefs, and answer a seven-item democratic-constraint battery. They then complete an unpaid guided

practice block covering the constrained contribution decision, executive taxation and transfer, the ballot, report prices, and random-round payment. Three mandatory comprehension questions must be answered correctly before the paid task begins.

Participants receive neutrally titled report cards about ordinary capacity, the budget and service environment, collective constraints, executive implementation, oversight, and the institutional trade-off. Opening a report reveals its full content and deducts four points from the information endowment. The Wave-1 reports establish a common low-capacity environment: ordinary provision has multiplier 1.50, whereas executive implementation has multiplier 2.50.

Participants then play ten paid rounds. Every round follows the same sequence: secret institutional ballot; majority rule; contribution decisions under D or executive decisions under A; simultaneous outcome feedback; rematching. After Round 10, capacity and risk beliefs and the constraint battery are measured again. The software records the final ballot, the executive-vote share in Rounds 8–10, and all round-level behavior. One Wave-1 round is randomly chosen for payment.

3.4 Wave 2: memory, structural change, and reversal

At the second visit, participants first complete the memory module without access to prior materials. The module contains free recall, recognition of Wave-1 capacity, recall of the participant’s own final Wave-1 ballot, and a 0–100 prior probability that ordinary capacity has recovered.

The same matching pools are then exposed to their randomly assigned Wave-2 environment. Under *recovery*, ordinary capacity rises from 1.50 to 2.50. Under *persistence*, it remains 1.50. Executive productivity, executive authority, and the maximum transfer remain constant. Six neutrally titled reports cover the same topic structure in both conditions; only purchased report content reveals whether recovery occurred. Immediately after search, participants state a new recovery probability and a capacity assessment.

Participants play ten additional rounds under the Wave-2 parameters. The same ballot and payoff structure makes behavior directly comparable across waves. After the final round, the capacity and risk measures and constraint battery are repeated. The final Wave-2 ballot is the behavioral endpoint. One Wave-2 round is randomly paid, and the software adds the unspent information endowment, a 25-point second-visit completion bonus, the scored-belief bonus, and the participation payment.

4 Learning Across Repeated Play: Design Safeguards

Repeated play is an advantage only if procedural learning is not confused with the substantive learning of interest. The design therefore anticipates six implementation concerns.

Procedural learning. Participants may initially misunderstand payoff rules and improve merely through practice. A guided unpaid block and mandatory comprehension checks move rule learning before the paid data. Round 1 remains available for diagnostics, while the preregistered endpoint uses Round 10 and the late-round mean uses Rounds 8–10.

Reputation and reciprocity. Fixed five-person groups would allow personal histories to drive votes and contributions. Anonymous rematching within ten-person pools breaks stable bilateral relationships while keeping treatment exposure and synchronized interaction feasible. Participants never receive persistent identities.

Executive-role learning. Subjects could support delegation because they expect to become executive. Executive assignment is balanced among the least-served eligible members, preventing a few participants from monopolizing the role. Analyses will nevertheless control descriptively for prior executive service and realized rents.

Endogenous institutional experience. Groups that never elect one procedure receive less experience with it. Instructions and practice expose everyone to both payoff rules, costly reports describe both institutions, and realized experience is recorded. Treatment effects will be reported by prior exposure, but the primary intention-to-treat contrast will not condition on a post-treatment institutional path.

Wealth, fatigue, and end-game effects. Only one randomly selected round per wave determines game earnings, so accumulated wealth from early rounds does not mechanically alter later budgets. Wave instructions explain that each round has an equal payment chance. The ten-round horizon is disclosed and identical across conditions. Page timers, decision times, and timeouts identify fatigue or end-game behavior.

Cross-wave comparability. The institution labels, executive rules, endowments, sequence, pool structure, and number of rounds are held constant. The recovery treatment changes only ordinary productivity and the reports that accurately describe it. A persistence group identifies change caused by returning to the lab, memory decay, or additional play.

These safeguards do not attempt to eliminate all learning. The study requires participants to learn from consequences. They instead separate theoretically meaningful learning about institutional performance from avoidable learning about the interface or accidental partners.

5 Outcomes, Process Measures, and Estimands

Let $V_{iwr} \in \{D, A\}$ be participant i 's ballot in wave w and round r . Define Wave-1 executive endorsement as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = A\}. \quad (4)$$

For the theoretically relevant subgroup with $A_{i1} = 1$, democratic reversal is

$$R_i = \mathbb{1}\{V_{i,2,10} = D\}. \quad (5)$$

Let $T_g = 1$ indicate that matching pool g received structural recovery. The primary causal estimand is

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \quad (6)$$

estimated as an intention-to-treat contrast with standard errors clustered by the ten-person randomization pool. Because A_{i1} is measured before treatment disclosure, defining this subgroup does not condition on a post-treatment variable. The primary model will include prespecified baseline covariates only for precision; an unadjusted difference in means will also be reported.

Three secondary behavioral outcomes describe robustness: (1) the change in executive-vote share from Wave-1 Rounds 8–10 to Wave-2 Rounds 8–10; (2) the full round-by-treatment trajectory; and (3) group-level selection of each institution. Contribution, imposed tax, private transfer, public-account size, and realized payoff explain the strategic environment in which ballots change.

Process outcomes occur at defined points in the sequence: baseline, post-Wave 1, Wave-2 pre-information memory, immediate post-search beliefs, and post-Wave 2. They include perceived

capacity, perceived risk to constraints, the seven-item constraint index, free recall, recognition accuracy, own-vote recall, report openings and order, search time, information expenditure, and probabilistic recovery beliefs. The order of reports and battery items is randomized once per participant and then stored, preventing re-randomization on refresh.

Belief accuracy will be evaluated against assigned capacity status. Search is costly by design, but it remains a choice. Accordingly, any decomposition through search or belief change will be labeled exploratory unless supported by a preregistered causal mediation strategy. The clean confirmatory claim is the pool-randomized effect of recovery on reversal.

6 Randomization, Sample, and Power

The planned sample is 400 Wave-1 participants organized into 40 ten-person matching pools. Pools are assigned in approximately equal numbers to recovery and persistence before Wave 1 but treatment is concealed until Wave 2. With 10 percent attrition, the expected returning sample is approximately 360 participants. Recruitment should overbook each synchronized cohort and maintain a standby list because interaction requires complete five-person groups.

Power depends on three quantities that cannot responsibly be inferred from the twenty repeated decisions: the proportion endorsing A at the Wave-1 endpoint, the persistence-group reversal rate, and the intrapool correlation. A pilot will estimate these values, after which power will be evaluated by simulation at the pool level under the planned assignment and attrition process. Repeated rounds increase measurement reliability and identify trajectories; they do not turn 400 participants or 40 randomized pools into 8,000 independent observations.

Randomization code assigns whole ten-person pools and approximately balances the number of pools across conditions. Production sessions should contain a multiple of ten participants. If several laboratory rooms run concurrently, randomization and matching remain within the configured cohort so that participants never interact across inconsistent treatment states.

7 Payments and Laboratory Operations

At EUR 0.20 per point, the implemented parameters are designed to yield an expected total payment of roughly EUR 28–32, including the EUR 10 participation payment, depending on play and information purchases. Final budgeting will use pilot distributions rather than theoretical maxima. All payoff components are displayed separately at completion: selected Wave-1 round, selected Wave-2 round, unspent information points, completion bonus, scored-belief bonus, point total, performance payment, and participation payment.

Wave 1 and Wave 2 are run as synchronized sessions rather than as short individual drop-ins. The second visit is scheduled at the first visit and reminder messages are sent without revealing treatment. A meaningful second-visit completion bonus reduces attrition. Participant codes, not names, link the waves.

The operational protocol requires complete groups. Standby participants may enter only before the first paid round. Once play begins, no replacement subject inherits another participant’s history. Every strategic decision has a 90-second limit. If a participant times out, the software records the timeout and applies a neutral, prespecified default so that group members can continue. The session log must document late arrival, browser failure, early departure, manual intervention, and payment exceptions. Exclusion rules should be preregistered and should never depend on treatment-consistent behavior.

Before launch, two pilots are required. A technical pilot should use at least ten simultaneous clients and both treatment branches to stress wait pages, rematching, refresh persistence, timeouts, and payment calculations. A behavioral pilot should estimate duration, comprehension, Wave-1 delegation frequency, information demand, executive extraction, attrition risk, and payoff distributions. Parameters should be frozen after that pilot and before confirmatory registration.

8 Ethics and Interpretation

The experiment exposes participants to a setting in which concentrated authority can be materially effective. That feature is necessary to study conditional tolerance, but it creates an obligation to avoid deception and moral labeling. Instructions therefore use neutral procedure labels, disclose all payoff rules, and accurately explain executive discretion. Treatment-specific reports are true descriptions of the assigned game environment.

Consent explains the two-visit structure, interaction, anonymity, payment rule, data linkage, and right to withdraw. The debrief explains the democratic interpretation only after all incentivized decisions, clarifies that individual choices are not diagnoses of personal authoritarianism, and provides researcher and ethics-board contacts. Data exports should replace participant labels with study IDs and store the linkage key separately.

The interpretation will remain narrow. Choosing executive delegation in a stylized public-good environment is evidence about behavior under institutional trade-offs, not a complete measure of democratic commitment. The strongest contribution comes from the randomized change in those trade-offs and the observed within-person return to constraints.

9 Planned Deliverables and Decision Rule

The confirmatory package will contain the theory, hypotheses, full instructions, treatment reports, oTree code, randomization procedure, exclusion rules, primary estimand, and pool-level power simulation. The principal conclusion will follow a simple rule: evidence for structural democratic reversal requires a positive, substantively meaningful, and statistically distinguishable recovery effect on R_i among Wave-1 executive endorsers. Round-level trajectories, beliefs, memory, information search, and allocations will be used to assess whether the observed endpoint follows the proposed learning mechanism and to identify rival explanations.

This design preserves the original project’s core ambition—observing people learn and then potentially unlearn tolerance for unconstrained authority—while giving that learning a strategic and distributive foundation. Subjects vote over rules that govern other subjects, experience the consequences of those rules, decide whether evidence is worth buying, and return one week later to make the choice again under an experimentally altered structure.

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